

Himadhara Puthran S O

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Research Interests

Cybersecurity, Information Security, Cyber Forensics, Hardware Security, Embedded Systems, IoT Security, AIoT, Telecom Security, Linux, Reverse Engineering, Blockchain Technology, Cloud Computing and Quantum Computing

Education

M.Tech, ER & DCI-IT, C-DAC, Trivandrum Oct 2022 – May 2024

- CGPA: 8.64/10 (86.4%) First Class with Distinction
- **Cyber Forensics and Information Security**
- **Advisor:** Renin Roy K
- **Thesis:** SIM & eSIM Forensics for IoT and M2M Device
- **Coursework:** Information Security, Cyber Forensics, File System Forensics, Network Security, Advanced Machine Learning, Ethical Hacking, Cloud Computing

B.Tech, Trinity College of Engineering, Trivandrum June 2016 – Sept 2020

- CGPA: 7.31/10 (73.1%) First Class
- **Electronics and Communication**
- **Advisor:** Blesson Mathews Luke
- **Thesis:** Automatic Transformer Control of Low Tension Circuit Breaker (LTCB) Switch
- **Coursework:** Electronic Circuits, Digital Communication, Antenna and Wave Propagation, Microprocessor and Microcontroller, Embedded Systems, Computer Communication, Advanced Communication Systems, Computer Vision

Experience

Multimedia Forensic Expert (Telecom), National Forensic Sciences University (NFSU), Gandhinagar, Gujarat Nov 2025 – March 2026

- Telecom fraud mapping, detection and analysis
- Research on Deepfakes, PBX fraud, SIM box, Voice cloning
- Telecom forensic frame work development and SOP implementation
- Digital and multimedia artifacts collection and tracing the evidence
- Audio, Video, Image Forensics, and AI assisted examination
- Deepfake detection and analysis (Audio, Video and Image)
- NLP and LLM based fraud detection and multimedia analysis

Cyber-Forensic Researcher (Intern), C-DAC, Thiruvananthapuram, Kerala Mar 2024 – July 2024

- Conducted in-depth research in the field of IoT Forensics
- Implemented a forensic framework for eSIM Integrated M2M and IoT Device
- Developed methodologies for Linux memory forensics analysis and Implemented various investigative strategies for small-scale IoT devices

Cyber-Security Researcher (Intern), C-DAC, Thiruvananthapuram, Kerala May 2023 – June 2023

- Carried out research in embedded systems and hardware security
- Conducted research on the CAN protocol and PKI X.509 Integration in Embedded Devices
- Performed reverse engineering of the CAN protocol in FDU.

Electronics Engineer (Apprenticeship), VSSC, ISRO, Thiruvananthapuram, Kerala June 2022 – Oct 2022

- Worked in the Avionics Department, collaborated with different projects.
- Projects: BioSensor system, and Programmable QCCVA based on PIC.
- Involved in different sensors and data acquisition techniques.

Cybersecurity EdTech Researcher, Adcy, Thiruvananthapuram, Kerala Feb 2021 – Dec 2021

- Conducted various research activities in cybersecurity domain
- Collaborated on creating advanced cybersecurity content for PG Diploma program.
- Performed various cyber attacks and cybersecurity practices for educational purposes.
- Familiarized with server deployment and network device management.
- Volunteered for various cybersecurity workshop programs.

Publications

Novel eSIM Forensic Framework for eSIM-Enabled M2M and IoT Devices July 2025
Himadhara Puthran S O, Amala R, Renin Roy K
DOI: 10.1109/ICCC64910.2025.11077302 [IEEE Conference](#)

IoT Trace X: A Forensic Framework for IoT, Embedded, and Smart Devices Accepted 2026
Himadhara Puthran S O, Krishnalal K K, Renin Roy K

Projects

SIM & eSIM Forensics for IoT and M2M Devices 2024

- Project aims to conduct, research study and effective framework for eSIM forensics analysis in the IoT/M2M and Embedded/Telematics devices. (Domain: Digital Forensics) (Role: Researcher)

Advanced Security Using PKI in DNS 2023

- Objective of the project is to provide advanced security in DNS. Primarily focusing on DNS Hardening by adding PKI and DNSSEC technology into DNS will ensure secure communication with client and server. (Domain: Cybersecurity) (Role: Researcher)

Security Audit in Fuel Dispenser System 2023

- PKI (X.509) digital certificate extraction and verification. The research study carried out in Fuel Dispenser Unit (FDU). (Domain: Embedded System) (Role: Researcher)

Bio-Sensor System & Programmable QCCVA Using PIC 2022

- Project is part of experimental setup and data acquisition and process the data from different sensors and module. (Domain: Space Technology) (Role: Researcher)

Website Hosting in Public Cloud Environment 2022

- Objective of this project is to host a static/dynamic website using a public cloud environment like Azure or AWS. Project is effective against heavy internet traffic & load. (Domain: Cloud Computing) (Role: Researcher)

Awards and Achievements

- Presented research paper in IEEE - International Conference on Control, Communication and Computing (ICCC) (Year: 2025)
- Contributed AI for Impact APAC Hackathon and submitted innovative solutions that promote environmental sustainability through the use of the AirOzia interactive agent. (Year: 2025)
- Linux Foundation LiFT Scholar Winner (Year: 2023)
- Guest Speaker at The Hacker Meetup Event (THM) Ahmedabad Chapter (Year: 2025)

Professional Service

Research Paper Writing and Reviewing

- Conducted quality research in Electronics, Cybersecurity and Digital Forensics. And Familiar with advanced system design, implementation, and result analysis using manual intervention and AI/ML
- Technical Reviewer, IEEE International Conference on Emerging Trends in Information, Communication & Systems (ICETICS) | 2026

Teaching and Mentoring

- Taught Cybersecurity subjects in different batches
- Academic presentations, training, and counseling

Volunteering Activities

- Conference talks in Cybersecurity and Internet safety
- Volunteer for social humanitarian activities and member of volunteer for India

Certifications

- Certified Linux System Administration (LFS207): Linux Foundation
- Certified Cloud Computing Engineer (NSQF Level 5): NIELIT
- Cybersecurity Professional Certification: Google and Coursera
- Certification in Cloud Computing and Data Science for Engineers (Elite): NPTEL

Skills

Cybersecurity Skills

- Linux and Windows System Security, Network Security, and Kali Linux
- Ethical Hacking, SIEM Tools, IoT Security, Bash Scripting, Cloud Management and Reverse Engineering
- Risk Management, Security Audit, OSINT Framework, SIEM, IDPS, Proxy Server
- OWASP Top 10, MITRE ATT&CK, NIST 800, NIST CSF and RMF, ISO 27001, VAPT and PTES
- Tools: SNORT, Wireshark, Hashcat, John the Ripper, Burp Suite, Nmap, Maltego, Jira, Bugzilla, Shodan, Metasploit Framework, Veil Framework, Empire Project, SET Toolkit

Digital Forensic Skills

- Windows and Linux Forensic, IoT Forensic, Embedded Device Forensic, Smartwatch Forensic
- Mobile Forensic, Digital Vehicle Forensic, Memory Forensics, SIM and eSIM Forensic
- Multimedia Forensic, Chip-Based Data Acquisition, Audio Spectrum and Spectrogram Analysis
- SWGDE Manuals, AES Audio Manual, ISO 27037:2012, ISO 27042:2015, ISO 27041:2015, ISO 17025
- Tools: Axon Investigate, Foclar Mandet, TrueImager, Amped Five, iZotope, Diamond Cut Audio, Praat, FFmpeg, MediaInfo, LiME, AVML, Volatility, EnCase, FTK, UFED and Autopsy

Computer Skills

- Operating Systems: Linux and Windows System Administration, and Android
- Programming Languages: Python, HTML, and Bash Scripting
- Protocols: TCP/IP, Serial I/O (CAN, RS232/485 & USB), RJ45
- Virtualization: VMware, VirtualBox, KVM, Xen, Hyper-V, Docker, Podman, Vagrant, and Terraform
- Linux and Windows System Administration: User & Group management, server deployment, etc.
- Applications: Kali Linux, Black Arch, Recon tools, OSINT tools, DVWA, Network simulation (GNS3, CISCO Packet Tracer, etc.), Forensics tools (Trueback, Karen's Hasher, and Encase, etc.)
- Hardware: IoT Module Testing, IoT Fuzzing and Sandboxing and Reverse Engineering
- Electronics: Circuit Design, Device Calibration, Testing, and Verification.